

Program: Center:

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Let's learn about programming languages.



Can Do Statements:

- I can read about programming languages.

- I can explain the importance of language programming in a writing activity.

- I can write about programming languages.

- Where can you find programming languages?
- Do you know how to design a web page?
- What do you need to create one?



Let's Explore!

Finding out* information about programming languages.

*"Find out" is a phrasal verb that means to discover something by searching or through experience.



Instructions: Read the text below.

Luis: Hi, did you read the book about programming languages?

Laura: Actually, I'm reading about that. But it's not very clear to me.

Luis: Well, as you know any form of communication requires a specific code. Programming language is the formal language people use to give instructions to a machine. Through that, people can create programs to control machine's behavior or to express algorithms. Besides, thanks to that, people designed programmable machines, digital computers, software, and many other technological devices we use nowadays.

Laura: Hum, I understand.

Luis: It's necessary that you learn how to write taking into account language programming. As you know codes are inflexible, so, you require typing exactly what the computer can recognize. This is the way you transfer information to a machine. Later, I'll show you an example I wrote.

Laura: Ok, thanks. I also found in this article that language programming has two components: syntax and semantics.

Luis: You're right! Those are two very essential components. So, the first one is syntax, this refers to a set of rules, that tells you how to put words and phrases together in order to form good sentences. But, specifically in computer language, it means the spelling and grammar of a programming language. Through that, you can describe possible symbol combinations. Additionally, you use sequences of organized text, for example, words, numbers, and punctuation; and as you can see this is similar to a natural language.

Laura: And what about semantics?

Luis: Well, I wrote here, an example of language programming. In this, you can identify the combination of symbols in which together they show a specific meaning, so, you call this: semantics. All those codes that you see have a special meaning; that is the case of any simple or complex computer program.



**Let's Practice!**

What is programming language?

Instructions: Choose the correct word to complete the sentence.

1. A (programming / programmable) language is the formal language people use to give instructions to a machine.
2. People (can / can't) create programs to manage its behavior or to express algorithms.
3. People (designed / design) programmable machines and digital computers.
4. Language programming (has / have) two components: syntax and semantics.

**Let's Practice!**

We can use algorithms to create instructions in that machine.

*"Algorithm" is a set of rules that express how to solve a problem.

Instructions: Complete the sentences by writing its corresponding word. Use capital letter where necessary. Use: programmable machines, codes are inflexible, specific language.

1. People designed _____, digital computers and software.
2. As you know _____, so, you require typing exactly what the computer recognizes.
3. It's necessary that you learn how to write taking into account _____ programming.



Extension Activity

Explaining the importance of programming languages.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- With the following adjectives: Specified, programmable, declarative, and dominant. Write down sentences about programming languages.
- Create a brochure, cut out and paste some pictures that can show examples about programming languages.
- Share your work to your classmates and check pronunciation with your instructor.

Self-Assessment

Yes

No

- I read about programming languages. ☐ Yes ☐ No
- I explain the importance of language programming in a writing activity. ☐ Yes ☐ No
- I write about programming languages. ☐ Yes ☐ No



Target Words

- **type** (verb): to write something on an electronic device by using a keyboard
e.g. As you know codes are inflexible, so, you require typing exactly what the computer can recognize.
- **take into account** (expression): similar to consider
e.g. It's necessary that you learn how to write taking into account language programming.
- **programmable** (adjective): something which is possible to program
e.g. Besides, thanks to that, people designed programmable machines, digital computers, software, and many other technological devices we use nowadays.
- **behavior** (noun): the way of acting
e.g. Through that, people can create programs to control machine's behavior, or to express algorithms.
- **inflexible** (adjective): not easy to change
e.g. As you know codes are inflexible, so, you require typing exactly what the computer can recognize.
- **essential** (adjective): similar to important or necessary
e.g. You're right! Those are two very essential components.



Learning Points

Action Verbs Used With Different Tenses To Talk About Programming Languages.

You can use **action verbs** to show that the subject performs some kind of action.

You can use them with different tenses, for example: present simple, present continuous, and past simple. Also you can use action verbs with the modal verbs such as can.

In English, you can find **state verbs** and **action verbs**. **State verbs** express an opinion, a feeling, or a state. Some verbs are: *like, dislike, love, agree, prefer, and own*. You don't usually use them in the present continuous, because they describe a state not an action in progress.

On the other hand, you can use the **action or dynamic verbs** to express an action. You can make different tenses with action verbs. Let's see some examples:

Present simple

You use the **present simple** to refer to routines, daily events, and things that usually happen or actions that you frequently do.

Example:

- He **works** with people that use programming language programs.

Present continuous

You use the **present continuous** to say that an action is happening at the moment.

Example:

- He **is working** with people that use programming language programs.

Past simple

You use the **past simple** to talk about a completed action in the past.

Example:

- He **worked** with people that use programming language programs.

Modal verbs: Can

You use the modal verb can to express different purposes: ability, possibility, or permission.

Example:

Can he **work** as an assistant using exploration of algorithms?





Learning Points

Present Simple			
	Affirmative	Negative	Questions
I/you/we/they	You send information about programming language programs.	We don't send information about programming language programs.	Do they get information about programming language programs?
He/she/it	He sells books about programming language programs.	She doesn't buy books about programming language programs.	Does she buy books about programming language programs?

Present continuous			
	Affirmative	Negative	Questions
I	I am selling a book that talks about the two components of programming language.	I am not buying a book that talks about the two components of programming language.	Am I studying about the two components of programming language?
you/we/they	You are selling new products related to the exploration of algorithms.	They aren't working in the office where technological devices are essential.	Are we buying products to support programming language programs?
he/she/it	She is working with exploration of algorithms.	He isn't studying about the components of programming language.	Is she writing a book about advances in programming language programs?



Learning Points

Past Simple			
	Affirmative	Negative	Questions
I/you/we/they He/she/it	You bought new technological devices that use programming language programs.	She didn't sell new products related to the exploration of algorithms.	Did they get products related to the exploration of algorithms?

Modal verbs: CAN			
	Affirmative	Negative	Questions
I/you/we/they He/she/it	You can buy new products related to programming language programs.	She can't sell new books related to the components of programming language.	Can we get related to the components of programming language?

Examples

- *Did you read the book about programming languages?*
- *I'm reading about that. But it's not very clear to me.*
- *Well, as you know any form of communication requires a specific code.*
- *Programming language is the formal language people use to give instructions to a machine.*
- *I also found in this article that language programming has two components: syntax and semantics.*





Learning Points

Adjectives Used in Programming.

You can use **adjectives** to give descriptions about programming or any other noun, for example, to describe people, animals, and things.

Remember, you need to put adjectives in front of nouns.

Example:

- A **big** computer.

Also, **adjectives** don't change. In other words, for singular and plural nouns the **adjective** stays the same.

Example:

- This is an **excellent** machine.
- These are two **excellent** machines.

You can use more than one **adjective** in front of a noun. In that case you place opinion adjectives (example: nice, ugly, terrible) before the fact adjectives (example: big, Chinese)

Example:

- This is a **terrible Chinese** computer.

Examples

- *You're right! Those are two very essential components.*
- *Additionally, you use sequences of organized text, for example, words, numbers, and punctuation; and as you can see this is similar to a natural language.*
- *All those codes that you see have a special meaning; that is the case of any simple or complex computer program.*